Section 1: Identification

- **Product Name:** Mobile AI
- **Models Covered:** Mobile AI (NO Laptop), Mobile AI (w/ Laptop)
- **Product Type:** Autonomous robotic system
- **Manufacturer / Supplier:** Trossen Robotics
- **Recommended Use:** Research, development, and educational applications
- **Address:** 2854 Hitchcock Ave, Downers Grove, IL, 60515, United States
- **Phone:** +1(630)437-5128
- **Emergency Phone Number:** CHEMTREC (24-hour): +1(800)424-9300
- **Email:** sales@trossenrobotics.com
- **Support:** https://www.trossenrobotics.com/support
- **Revision Date:** 2026-02-04

Section 2: Hazards Identification

2.1 Classification of the Substance or Mixture

This product is an article and is not classified as hazardous according to the Globally Harmonized System (GHS). No hazardous substances are released under normal conditions of use.

2.2 GHS Label Elements

Signal Word: None
Hazard Pictograms: None
Hazard Statements: None
Precautionary Statements: None

2.3 Other Hazards

Although not classified as hazardous under GHS, the product may present **non-chemical hazards** associated with its electromechanical operation, installation, servicing, misuse, or abnormal conditions. These hazards are not subject to GHS classification but are identified for safety awareness and risk management purposes.

Potential non-chemical hazards include:

- **Electrical Shock:** Risk of electric shock from power supplies, wiring, or internal components if the product is improperly installed, damaged, or serviced while energized.
- **Mechanical Hazards / Moving Parts:** Risk of injury from moving components, including pinch points, rotating elements, or articulated mechanisms during operation or maintenance.

- **Unexpected Motion (Autonomy):** Autonomous or semi-autonomous operation may result in unanticipated movement if safety interlocks, sensors, or software controls are bypassed, misconfigured, or fail.
- **Battery Hazards:** Integrated or external batteries may present risks of overheating, fire, thermal runaway, or electrolyte leakage if damaged, improperly charged, or exposed to extreme temperatures.
- **Optical / Laser Sensor Exposure:** All models incorporate camera or optical sensing devices. Direct exposure does not present a chemical hazard; however, removal of protective housings or misuse may pose minor optical exposure risks.
- **Hot Surfaces:** Some components may become warm or hot during extended operation or charging, presenting a burn hazard upon contact.

These hazards are mitigated through product design, protective enclosures, software controls, labeling, and operating instructions when the product is used as intended.

2.4 Hazards Not Otherwise Classified (HNOC)

None known.

Section 3: Composition / Information on Ingredients

3.1 Substances

Not applicable.

3.2 Mixtures

Not applicable.

3.3 Articles

This product is an **assembled electromechanical article** and does **not release hazardous chemicals** under normal conditions of use.

The product consists of mechanical structures, electronic assemblies, wiring, sensors, power supplies, and accessories that are **fully enclosed or integrated** during normal operation.

Components of Potential Regulatory Interest

Lithium-ion Batteries

Some configurations of this product incorporate rechargeable lithium-based battery systems integrated within power stations, mobile bases, controllers, laptops, or accessories. Battery chemistry may vary by configuration and may include lithium-ion (Li-ion) or lithium iron phosphate ($LiFePO_4$) technologies.

All batteries supplied with the product are sealed, contained units. No exposure to hazardous substances is expected during normal conditions of use. Potential hazardous exposure may occur only if batteries are damaged, improperly handled, overcharged, or involved in fire or thermal events.

Lithium-ion batteries are regulated separately for transport, handling, and disposal (see Sections 7, 13, and 14).

Other Components

This product may include, but is not limited to:

- Electronic circuit boards and controllers
- Power supplies and charging systems
- Computing devices (e.g., laptop computers with internal batteries)
- Sensors, cameras, and networking equipment
- Structural metal and polymer components
- Cables, connectors, and fasteners

These components do **not present chemical hazards** under normal conditions of use and are not classified as hazardous under GHS.

Additional Information

No components are present above reportable thresholds requiring disclosure under GHS, OSHA Hazard Communication, or equivalent international regulations.

Section 4: First Aid Measures

4.1 Description of First-Aid Measures

General Advice:

No chemical exposure, inhalation, eye contact, or ingestion is expected under normal conditions of use. If an accident occurs, provide first aid appropriate to the nature of the injury and seek medical attention as indicated.

Inhalation:

If fumes or vapors are inhaled due to battery damage, overheating, or fire, move the affected person to fresh air immediately. Seek medical attention if symptoms persist.

Skin Contact:

In the event of contact with battery electrolyte due to battery rupture or leakage, immediately remove contaminated clothing and rinse affected skin with plenty of water for at least 15 minutes. Seek medical attention.

Eye Contact:

If debris, particulate matter, or battery electrolyte contacts the eyes, flush immediately with clean water for at least 15 minutes. Seek medical attention.

Ingestion:

Ingestion of product components is not expected. If battery contents are ingested due to catastrophic failure, rinse mouth with water (do not induce vomiting) and seek immediate medical attention.

4.2 Most Important Symptoms and Effects

- Electrical shock
- Mechanical injury
- Thermal burns
- Eye or skin irritation from battery electrolyte exposure (in abnormal conditions)

4.3 Indication of Immediate Medical Attention

Immediate medical attention is required in cases of:

- Electrical shock
- Significant mechanical injury
- Suspected battery electrolyte exposure
- Smoke or fume inhalation resulting from battery overheating or fire

Section 5: Fire-Fighting Measures

5.1 Suitable Extinguishing Media

- Carbon dioxide (CO_2)
- Dry chemical powder
- Fire-fighting foam
- Water spray or fog (for cooling and fire control)

5.2 Specific Hazards Arising from the Product

- Combustion of plastic, polymer, and electronic components may release hazardous fumes

- Lithium battery systems may undergo thermal runaway if exposed to fire or excessive heat
- Toxic and corrosive gases may be emitted during battery combustion

5.3 Advice for Firefighters

- Wear self-contained breathing apparatus (SCBA) and full protective fire-fighting gear
- Use caution when approaching energized or battery-powered equipment
- If safe to do so, disconnect external power sources and isolate battery systems
- Cool batteries exposed to fire using copious amounts of water to prevent re-ignition

Section 6: Accidental Release Measures

6.1 Personal Precautions

No chemical release is expected under normal use

In the event of battery damage, leakage, overheating, or smoke emission:

- Evacuate the area
- Avoid contact with leaking materials
- Do not inhale fumes
- Use appropriate PPE (gloves, eye protection, respiratory protection if necessary)

6.2 Environmental Precautions

- Prevent battery materials or electrolyte from entering drains, soil, or waterways
- Contain and collect released materials in accordance with local regulations

6.3 Methods for Cleanup

- Do not handle damaged batteries with bare hands
- Place damaged batteries in non-combustible, sealed containers
- Follow manufacturer guidance and local regulations for disposal or recycling

Section 7: Handling and Storage

7.1 Handling

- Operate the product only in accordance with manufacturer instructions
- Disconnect power before servicing or maintenance
- Do not open sealed battery systems or internal electronic enclosures
- Avoid crushing, puncturing, or subjecting the product or batteries to mechanical shock
- Use proper lifting techniques when transporting the system

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7.2 Storage

- Store in a dry, well-ventilated area
- Avoid exposure to moisture, corrosive environments, or direct sunlight
- Store within recommended temperature ranges specified by the manufacturer
- For systems containing lithium batteries or laptops with internal batteries:
    - Do not store near heat sources
    - Follow battery manufacturer storage recommendations
    - Avoid long-term storage at full charge unless specified otherwise

Section 8: Exposure Controls / Personal Protection

8.1 Control Parameters

- Occupational exposure limits: Not established / Not applicable
- No chemical exposure is expected during normal operation

8.2 Exposure Controls

Engineering Controls:

- Protective enclosures
- Emergency stop functionality
- Software interlocks and safety controls
- Electrical insulation and grounding

Personal Protective Equipment (PPE):

- **Eye Protection:** Safety glasses during servicing or maintenance
- **Hand Protection:** Protective gloves when handling components or damaged batteries
- **Skin Protection:** Not required under normal use
- **Respiratory Protection:** Not required under normal use; required if fumes are present due to battery failure or fire

Hygiene Measures

- Wash hands after servicing
- Do not eat, drink, or smoke while performing maintenance

Section 9: Physical and Chemical Properties

9.1 Information on Basic Physical and Chemical Properties

- **Physical State:** Solid article
- **Appearance:** As manufactured (assembled robotic system)
- **Odor:** None
- **pH:** Not applicable
- **Melting / Freezing Point:** Not applicable
- **Boiling Point:** Not applicable
- **Flash Point:** Not applicable
- **Evaporation Rate:** Not applicable
- **Flammability (Solid/Gas):** Not flammable as supplied
- **Upper/Lower Flammability Limits:** Not applicable
- **Vapor Pressure:** Not applicable
- **Vapor Density:** Not applicable
- **Relative Density:** Not applicable
- **Solubility:** Insoluble in water
- **Auto-ignition Temperature:** Not applicable
- **Decomposition Temperature:** Not determined

9.2 Other Information

- Battery systems may generate heat during operation and charging

Section 10: Stability and Reactivity

10.1 Reactivity

No hazardous chemical reactivity is expected under normal conditions of use.

10.2 Chemical Stability

Stable under normal operating, handling, and storage conditions.

10.3 Possibility of Hazardous Reactions

No hazardous reactions expected during normal use.
Lithium battery systems may present hazards if damaged, improperly charged, or exposed to fire.

10.4 Conditions to Avoid

- Water ingress into electrical systems
- Mechanical damage, crushing, or puncturing
- Exposure to temperatures outside manufacturer specifications
- External short-circuiting of battery systems

10.5 Incompatible Materials

None known under normal use.

10.6 Hazardous Decomposition Products

In the event of fire or battery failure:

- Toxic and corrosive gases
- Combustion byproducts of plastics and electronic components

Section 11: Toxicological Information

11.1 Information on Likely Routes of Exposure

- Inhalation: Not applicable under normal use
- Skin contact: Not applicable under normal use
- Eye contact: Not applicable under normal use
- Ingestion: Not applicable

11.2 Symptoms Related to Physical, Chemical, and Toxicological Characteristics

No toxicological effects are expected during normal operation.
Potential adverse effects may occur only under abnormal conditions, including:

- Battery electrolyte exposure
- Smoke or fume inhalation during battery overheating or fire

11.3 Information on Toxicological Effects

This product is an article and does not release hazardous substances under normal conditions.
Toxicological data for internal battery materials are available in manufacturer SDS documents upon request.

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Section 12: Ecological Information

12.1 Toxicity

No known environmental toxicity under normal conditions of use.

12.2 Persistence and Degradability

Not applicable for an assembled article.

12.3 Bioaccumulative Potential

Not applicable.

12.4 Mobility in Soil

Not applicable.

12.5 Results of PBT and vPvB Assessment

Not applicable.

12.6 Other Adverse Effects

Improper disposal of batteries or electronic components may cause environmental harm.

Section 13: Disposal Considerations

13.1 Waste Treatment Methods

- Dispose of electronic components in accordance with local, regional, and national e-waste regulations
- Do not dispose of lithium batteries in general waste streams
- Do not incinerate batteries
- Recycle batteries through approved recycling programs

Follow all applicable regulations, including WEEE, EPA, and equivalent international directives.

Section 14: Transport Information

This product is not regulated as a dangerous good when shipped as an assembled system with lithium-ion batteries contained in equipment (UN 3481).

Shipments must comply with applicable lithium battery transport regulations.

Lithium Battery Transport:

- **UN Number:** UN 3480 (Lithium-ion batteries)
- **UN Number:** UN 3481 (Lithium-ion batteries contained in or packed with equipment)
- **Transport Hazard Class:** 9
- **Packing Group:** Not applicable
- **Environmental Hazards:** None

Transport must comply with:

- IATA (Air)
- IMDG (Sea)
- ADR / DOT (Ground)

Section 15: Regulatory Information

15.1 Safety, Health, and Environmental Regulations

- **OSHA Hazard Communication Standard:** Not classified as hazardous
- **GHS:** Not classified
- **REACH:** Article; no SVHC above reporting thresholds
- **RoHS:** Compliant (where applicable)
- **California Proposition 65:** This product may contain trace substances subject to Prop 65 requirements.

This product complies with applicable electrical, EMC, and safety regulations where sold.

Section 16: Other Information

Additional Information

Certain configurations of this product incorporate third-party rechargeable lithium battery systems. Battery chemistry and form factor may vary by configuration and may include lithium-ion (Li-ion) and lithium iron phosphate ($LiFePO_4$) technologies integrated within power stations, mobile bases, controllers, or computing devices (e.g., laptop computers with internal batteries).

Applicable manufacturer Safety Data Sheets (SDS) for specific battery systems are available upon request and should be consulted for detailed chemical composition, toxicological, firefighting, and transport information.

Under normal conditions of use, these batteries are sealed units and do not release hazardous substances.

Referenced SDS

Manufacturer Safety Data Sheets for battery systems are available upon request.

SDS Metadata (Recommended for Audit Readiness)

- **SDS Preparation Date:** 2026-02-04
- **Revision Date:** 2026-02-04
- **Revision History:** Initial release

Disclaimer

This Safety Data Sheet is provided for informational purposes to support safe handling and use of the product. It does not replace product manuals, operating instructions, or applicable regulatory requirements.

Contact for SDS Information

Trossen Robotics
Email: sales@trossenrobotics.com
Support: https://www.trossenrobotics.com/support

